Zoo-511T

Principles Of Systematics (Objective)

VU Expert B.S Zoologists
(All Semesters)

CURRENT PAPER SPRING 2024

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“””Multiple Choice Questions”””

1. **Spider venom is used in the treatment of disease:** **Cancer**
2. **Which organ of arthropods is used to inject venom:** **Chelicerae**
3. **Which is not a component of silk:** **Phospholipid**
4. **Length of egg of green lacewing:** **Approximately 1/2 inch**
5. **Diploblastic organisms don't have:** **Mesoderm**
6. **Spiders have:** **4 pairs of eyes**
7. **Which is not a homologous structure:** **Wings of a bird**
8. **Color of compound eyes of whitefly:** **Red**
9. **Blue and yellow sticky traps:** **Thrips**
10. **Organisms that are morphologically similar but reproductively isolated:** **Sibling species**
11. **Scientific name of leopard shark:** **Triakis californica**
12. **Word taxis means:** **Arrangement**
13. **Measure of central tendency for normal data is mean and median for:** **Skewed data**
14. **Degrees of Freedom (DF) within group:** **Number of observations - number of groups**
15. **Degrees of Freedom (DF) between groups:** **Number of groups - 1**
16. **Our null hypothesis is rejected if calculated value is:** **Greater than the tabulated value**
17. **Accumulation curve is used to represent:** **cumulative number of species

18. **Which is not a part of reference:** **Place of collection**

19. **Evolution is a:** **Gradual and continuous process**

20. **Animal that kills and consumes other organisms:** **Predator**

21. **Species are the product of:** **Evolution**

22. **The first animal that moved from water to land:** **Likely amphibians**

23. **Which organisms maintain temperature by absorbing heat from the environment:** **Amphibians**

24. **Which organisms are exclusively marine:** **Echinoderms**

25. **A character which is consistently present in one group and absent in a related group should have** high weight**

26. **Complicated form of cluster sampling:** **Multistage sampling**

27. **Phenogram definition:** **True**

28. **Chelicerae comprises:** **Coxa, tibia, tarsus**

29. **No taxonomic study is completed until its:** **Results are published**

30. **Who is called the Darwin of the 20th century?** **Ernst Mayr**

31. **What does the term "phenon" refer to?** **A term used for the different forms or phenotypes that may occur within a single population**

32. **Why is fixation a crucial step in animal preservation?** **It halts decomposition and preserves the structure**

33. **Epigastric plates are on which body part?** **On the ventral side of abdomen**

34. **Evolutionary trees are constructed based on:** **Homology**

35. **Which of the following is used to study the evolutionary history of species:** **Phylogenetics**

36. **The term "phylogeny" refers to:** **Evolutionary relationships**

37. **Genetic drift has a larger effect in:** **Small populations**

38. **Which of the following is not a type of genetic drift:** **Natural selection**

39. **Homologous structures indicate:** **Common ancestry**

40. **Analogous structures arise due to:** **Convergent evolution**

41. **The study of fossils is called:** **Paleontology**

42. **The term "speciation" refers to:** **Formation of new species**

43. **Which of the following is an example of allopatric speciation:** **Geographically isolated**(Galapagos finch)

44. **Sympatric speciation occurs without:** **Geographic isolation**

45. **The process by which organisms better adapted to their environment tend to survive and produce more offspring is called:** **Natural selection**

46. **Genetic recombination occurs during:** **Sexual reproduction**

47. **Which of the following is not a mechanism of reproductive isolation:** **Gene flow**

48. **The Hardy-Weinberg principle describes:** **Genetic equilibrium**

49. **Which of the following is not a condition for Hardy-Weinberg equilibrium:** ***

50. **Which of the following is a prezygotic barrier:** ****

51. **Which of the following is a postzygotic barrier:** ****

52. **The study of the distribution of species and ecosystems in geographic space and through geological time is called:** **Biogeography**

53. **Which of the following is not a type of genetic variation:** **Homologous**

54. **Which concept emphasizes the importance of genetic differences in evolution:** **Population genetics**

55. **The process by which unrelated species evolve similar traits:** **Convergent evolution**

56. **The primary source of genetic variation in a population:** **Mutations**

57. **The evolutionary process by which new species arise from a single ancestral species:** **Speciation**

58. **The phenomenon where an organism's traits are more suited to a particular environment due to evolutionary pressures:** **Adaptation**

59. **The process by which traits are passed from parent to offspring:** **Inheritance**

60. **The study of how traits are inherited from one generation to the next:** **Genetics**

61. **The term used to describe the diversity of species within a particular habitat:** **Biodiversity**

62. **The type of evolution where two or more species become more dissimilar over time:** **Divergent evolution**

63. **The principle stating that only a small fraction of the genetic variation is due to selection:** **Neutral theory**

64. **The concept that species evolve in response to changes in their environment:** **Ecological evolution**

65. **The study of the relationships among different species:** **Systematics**

66. **The process of identifying and classifying organisms based on their characteristics:** **Taxonomy**

67. **The field that examines the evolutionary relationships among groups of organisms:** **Phylogenetics**

68. **The role of genetic mutations in evolution:** **Creating new alleles**

69. **The role of genetic recombination in evolution:** **Shuffling genetic material**

70. **The impact of gene flow on a population:** **Mixing genetic material**

71. **The role of genetic drift in small populations:** **Causing random changes in allele frequencies**

72. **The Hardy-Weinberg principle assumes no:** **

73. **The study of evolutionary development and diversification of a species:** **Evolutionary biology**

74. **The concept of "adaptive radiation" refers to:** **Rapid diversification of a species**

75. **The term "fitness" in evolutionary terms refers to:** **Reproductive success**

76. **The study of the fossil record to understand evolutionary history:** ****

77. **The theory proposing that evolution occurs in short, rapid bursts followed by long periods of stability:** **Punctuated equilibrium**

78. **Do not maintain their body temperature:** **Amphibians**

79. **Jumping spider belongs to which family:** **Salticidae**

80. **Spider venom is present in which part of the body:** **Fang**

81. **Z test is used to compare mean of sample when sample size is greater than:** **30**

82. **Pearson coefficient of correlation is used in:** **Parametric test**

83. **Outcomes of match is an example of:** **Binomial distribution**

84. **Commonly used measure of central tendency:** **Mean, Median, and Mode**

85. **ANOVA is:** **Analysis of variance**

86. **T-test sample size:** **30 or less than 30**

87. **Spiders have legs:** **8**

88. **The eggs are either laid singly or in small groups. Each is always found perched on the tip of a hairlike stalk that is about:** **1/2 inch long**

89. **Predators kill and consume other organisms:** **Prey**

90. **The amphibian was the first animal that moved from water to land.**

91. **Arthropods use to inject venom: **Fang****

92. **Major sucking pest of cotton: **Jassid****

93. **Which part of scorpion is called Post abdomen: **Tail****

94. **Mayr described the History of Taxonomy into: **4 periods****

95. **King parrot Eclectus roratus of Papuan region, the Male is: **Green with orange bill****

96. **Who gave the Typological species concept: **Aristotle and Plato****

97. **Classification of organisms on the basis of distribution and composition of chemical substances in them: **Molecular biology****

98. **Why predator attacks on small prey: **To catch them easily****

99. **Speciation in which a new species is formed from an isolated peripheral population: **Parapatric speciation****

100. **Small insect or insect in hard to reach place can be collected using which organ: **Aspirator****

101. **One-way ANOVA uses: **1 independent variable****

102. **Not a type of data distribution: **Analogous distribution****

103. **Not part of cephalothorax: **Brain****

104. **Exoskeleton of arthropod is made up of: **Chitin****

105. **Polyclave is based on a computer program.**

106. **Rana tigrina is: **Nocturnal****

107. **Restricted use of RFLF technique: **Radioisotope****

108. **Study of anatomy, morphology, and complete fur color: **Entire fluid preserve specimen****

109. **Social variation is seen in: **Honey bee****

110

. **Genetic variations cause: **Phenotypic variations****

111. **Polyphyletic groups are formed: **To study evolution****

112. **In Chordata: **Notochord is present****

113. **Study of similarity between butterfly and bird wings: **Comparative morphology****

114. **Species classification in Zootaxonomy was done by: **Artificial method****

115. **Who proposed species concept on the basis of similarity: **Aristotle****

116. **True about identical characters: **Homoplasy****

117. **Science of classification of living organisms is called: **Taxonomy****

118. **The major cause of extinction: **Habitat loss****

119. **First person to give the definition of species: **John Ray****

120. **Taxon-specific group which follows a single line of descent: **Monophyletic****

121. **In amphibians, which part of the body excretes CO2: **Skin****

122. **In amphibians, what function is performed by the skin: **Respiration****

123. **Complicated sampling: **Multistage****

124. **Measurement of similarity is called: **Phenogram****

125. **RFLP is: **Radioisotope technique****

126. **Study of similarity between butterfly and bird wings: **Comparative morphology****

127. **Molecular systematics classification: **On the basis of DNA sequence****

128. **Who introduced Taxonomy for the first time: **Carolus Linnaeus****

129. **Study of identical characters in different organisms: **Homoplasy****

130. **Monophyletic groups: **Single ancestor****

131. **Taxon-specific groups: **Polyphyletic****

132. **Length of green lacewing egg: **1/2 inch****

133. **Spider's venom gland is present in: **Fang****

134. **Polyphyletic taxon refers to: **Multiple ancestors****

135. **Chelicerae are found in: **Arthropods****